Study on extraction of flavonoids from black beans and antioxidant activity

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Abstract: Using black soybeans as materials, study on extraction of flavonoids from black soybeans and antioxidant activity. On the basis of single factor, optimum technological parameter is obtained by the response surface method optimization. The antioxidant activity in different concentration of black bean andcompared with Vc. The result showed that: the optimum extraction conditions were: solid-liquid 30:1,ethanol concentration 70%, temperature 60℃, the yield offlavonoids was 41.0223 mg/100g, attain 97.698%.The antioxidant ability of flavonoids is higher than that of Vc.

Key Words:lack beans; flavonoids; response surface; antioxidant activity

Introduction:

Black soybean is the seed of leguminous plants. Flavonoids, exist widely in the nature, have high physiological activity and medicinal value. Reports about study on the extraction of flavonoids from black soybean have not seen. This experiment use alcohol to extract flavonoids from black soybeans and study antioxidant activity in vitro,it can provide scientific basis for the produce of flavonoids in health products or drugs.

1. Meterial and Methods

1.1 Material and Equipments

Black beans: from Xin yinghe Supermarket; Rutin sample、Vc、DPPH Sigma company; Phosphate buffer、Ferrisulfas、Salicylic acid、potassium ferricyanide、Trichloroacetic acid、Iron trichloride all are analytical reagent.

Equipments：RJ-TDL-40C Low-speed large-capacity centrifuge, Wuxi Ruijiang Analytical instruments Co,Ltd. UV-1100 type UV spectrophotometer, Shanghai Mei Puda Co.Ltd. GZX-9246NVBE digital blast drying box, Shanghai Boxun Industrial Limited Medical Equipment Co. Ltd. F80 High-speed disintegrator,Jincheng Jinchengguosheng Co.Ltd .

1.2 Experimental Method

1.2.1Pretreatment of raw material

Soaked black beans in water 30min, artificially removed the peels, drying, pulverized and passed through 40 mesh sieved, kept in desiccator at normal temperature. With petroleum ether to skim the black soybeans.

1.2.2The preparation of standard curve rutin

To add 10 mg of rutin into 50 ml of volumetric flask,constant volume by 70% ethanol, separately to add 0、0.5、1.0、2.0、3.0、4.0ml into10 ml volumetric flask and added 0.4 ml 5%$NaNO_2$ solution, shook up ,Shake for 6min ,added 0.4 ml 10% nitri acid aluminum, Shake for 6min ,added 2ml 4% NaOH solution, Shake for 15min, constant volume by 70% ethanol, the absorbence value is determined at 510 nm.Parallel measurement 3 times, get standard curve and equation: y=0.0094x-0.0011(R^2=0.9996）y: Absorbance Value，x: the concentration of ruit Standard solution mg/mL。

1.2.3Determination of flavonoids content

Took 1g Samples added a certain amount of extracting agent ,abstracted a certain time, got fluid under test. According to the above procedure of drawing standard curve rutin and to the regression equation calculating the flavonoids content of in samples.

1.2.4The optimization of black beans flavonoids extraction

Based on different liquid ratio,ethanol concentration, extraction time and extraction temperature on the rate of black beans flavonoid extraction, with Design-Expert8.0 software on three of the important factors for response surface optimization analysis.

1.2.5Experiment of flavonoids antioxidant activity

1.2.5.1 Determination of the scavenging rate of •OH fress radicals

Added 2.5ml 6.0mmol/L Ferrisulfas solution, 2.5ml 6mmol/L H_2O_2 solution and 7.5ml 3.0mmol/L Salicylic acid solution in turn ,Shake for 20min in constant temperature water bath water pot at 37℃, the absorbence value is determined at 510 nm, saided by A_0. Addion flavonoids extraction and Vc at different concentration in the above tubes, kept warm 10 min,measured absorbency value in 510 nm,saided by A_1.The yied of flavonoid is:$P(\%)=(A_0-A_1)/A_0\times100\%$,$A_0$:The absorbency of liquids without samples ,A_1: The absorbency of samples.

1.2.5.2 Determination of the scavenging rate of DPPH • fress radicals

Added 2ml flavonoids extraction samples and Vc at different concentrations, then added 4ml 25μg/ml DPPH solution, Shake for 30min, the absorbence value is determined at 517 nm ,marked A_1. Rate of clearance:$P(\%)=(A_0-A_1)/A_0\times100\%$，$A_0$： The absorbency of liquids without samples A_1: The absorbency of samples.

1.2.5.3The determination of reducing power

Added 1ml flavonoids extraction samples and Vc at at different concentrations, then added 2.5ml pH6.6 Phosphate buffer solution and 2.5ml 1% potassium ferricyanide solution, in water-bath at 50℃, Shake for 20min.added 2.5ml 10% Trichloroacetic acid solution, draw 2.5ml supernatant fluid 2.5ml 0.1%$FeCl_3$ solution and 2.5ml distilled water, Shake for 10min, the absorbence value is determined at 700nm.

2 Results and Analysis

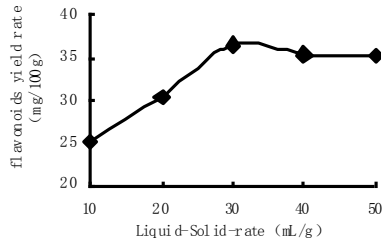
2.1 Results of single factor tests

2.1.1Effect of the yield of flavonoids on degreasing or not on natural black beans

The extraction rate of undegreasing black beans is 27.68mg/100g, the degreasing black beans is 35.13mg/100g.The lipid in black beans influence the solution of flavonoids in ethanol, so choose degreasing black beans.

2.1.2 Effect of liquid-solid ratio on yield of flavonoids

The results were showed in Fig1. From the Fig1, the extraction rate increases at first then decreases with liquid-solid ratio increasing, and reached maximum at 30ml/g .It's because flavonoids can better dissolve in extractant with ratio increasing, but the more ethanol extractant made the extraction rate decreased .So ,the optimum ratio of liquid-Solid is 30:1. From the Fig 2, the extraction rate increase with time increasing, 60min later flavonoids completely soluble in extractant, then the rate increased little with time continue increasing, select 60min as optimum extraction time.



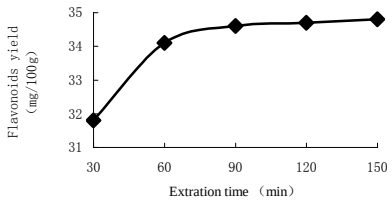
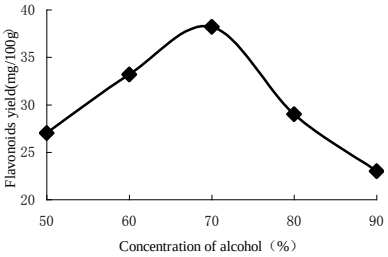


Fig1. Effect of liquid ratio on the extraction **Fig.2 Effect of extration time on extraction**

2.1.3Effect of ethanol concentration on yield of flavonoids

The results were showed in Fig 3.



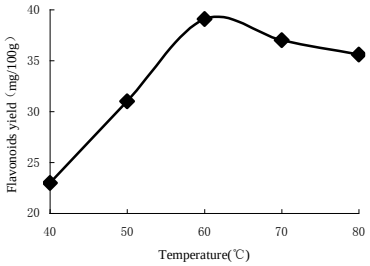


Fig.3 Effect of temperature on extraction **Fig.4 Effect of temperature on extraction**

From the Fig 3, with ethanol concentration increasing ,the flavonoids extraction rate increases first and reached maximum at 70%, then drops rapidly. Flavonoids are polar moleculeand soluble in polar solvents, the polar is different at different ethanol concentration, the solution is different. So select 70% ethanol extraction.

2.1.4Effect of extraction temperature on the yield of flavonoids

From the Fig 4 the flavonoids extraction rate increase first and reached maximum at 60℃,but drops when more than 60℃.The flavonoids are more easily dissolved in the solvent with increasing temperature, but it may be have harm on the structure of the flavonoids if temperature more than 60℃.So choose extraction temperature at 60℃.

2.2 **response surface optimization**

Choose liquid-solid ratio、extraction time and extraction temperature from the three single factors to optimize the main technological parameters by response surface .

2.2.1Results the response surface optimization

Table 1 Experiment assignment and the data

No	Response(mg/100g)				
	(X_1)	(X_2)	(X_3)	Test value	Predictive
1	-1	-1	0	36.43	36.27
2	1	-1	0	34.71	34.43
3	-1	1	0	32.76	33.04
4	1	0	1	34.11	34.27
5	-1	0	-1	34.01	33.99
6	1	0	-1	34.13	34.23
7	-1	0	1	36.11	36.00
8	1	0	1	35.14	35.16

9	0	-1	-1	35.12	35.30
10	0	1	-1	29.48	29.21
11	0	-1	1	32.10	32.37
12	0	1	1	35.25	35.07
13	0	0	0	41.90	41.93
14	0	0	0	42.20	41.93
15	0	0	0	41.92	41.93
16	0	0	0	41.71	41.93
17	0	0	0	41.97	41.93

Analyed datas with Design-Expert, got quadratic multinomial regression equations at many factors(liquid-solid ratio (X_1)、ethanol concentration(X_2) and extraction temperature(X_3)、interaction):$Y=41.93-0.15X_1-0.84X_2+0.73X_3+0.77X_1X_2-0.27XX_1X_3+2.2X_2X_3-2.78X_1^2-4.64X_2^2-4.30X_3^2$.

2.2.2 Regression model analysis of variance

Tab.2 Variance analysis of regression equation

Source	SS	DF	MS	F	P	significant
Model	255.58	9	28.40	352.62	<0.0001	**
Residual	0.56	7	0.081			
Lack of fit	0.44	3	0.15	4.79	0.0821	
Pure error	0.12	4	0.031			
total	256.14	16				

Through response surface analysis, the regression model of P values < 0.0001, extremely significant, so the model feasibility is good. Lack of fit P values is 0.0821 > 0.05, was not significant, showed the unknowns to test. $R^2 = 0.9950$, indicate that the better the fit of the model and the actual situation, can be used to predict the maximum yield of flavonoids and the best extraction technology of flavonoids.

2.2.3 Regression equation coefficient test

Table 3 Regression coefficients significant test

Model	Regression coefficient	df	$S_{\bar{x}}$	F	P	Significances
Constant	41.93	1	0.13			
X_1	0.19	1	0.19	2.31	0.1723	
X_2	5.17	1	5.71	70.93	< 0.0001	**
X_3	4.29	1	4.29	53.30	0.0002	**
X_1X_2	2.36	1	2.36	29.26	0.0010	**
X_1X_3	0.30	1	0.30	3.69	0.0963	
X_2X_3	19.32	1	19.32	239.85	<0.0001	**
X_1^2	32.61	1	32.61	404.87	<0.0001	**
X_2^2	90.76	1	90.76	1126.97	<0.0001	**
X_3 ²	77.77	1	77.77	965.71	<0.0001	**

From the table3: X_2 (ethanol concentration), X_3 (temperature) two P values were extremely significant, larger influence on the yield of flavonoid. In interaction, X_1X_2 (liquid ratio and ethanol concentration of interaction) and X_2X_3 (ethanol concentration and extraction temperature) P values reached extremely significant level, the biggest influence on flavonoid yield. Quadratic term P values were extremely significant, have great influence

on the yield of flavonoid

2.3 Results of antioxidant activity.

2.3.1 Determination of the scavenging rate of •OH fress radicals

From the Fig 5, scavenging rate of OH • of flavonoids and Vc were increasing at 7μg/ml～16μg/ml，but increased modestly.When it is more than 16μg/ml, increased significantly. In the experimental concentration range, 19μg/ml flavonoids scavenging of OH • is 43.65%,increases to the maximum.

2.3.2 Determination of the scavenging rate of DPPH • fress radicals

The results were showed in Fig 6

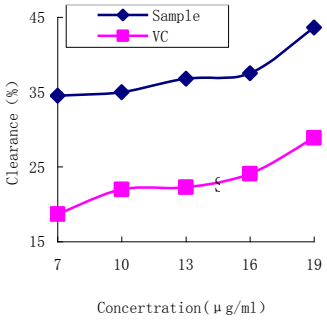


Fig.5 Ethanol concentration and temperature on the extraction flavonoids in the interactive effects

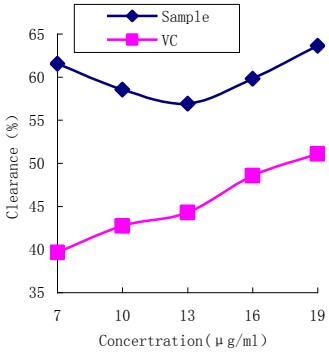


Fig 6.flavonoid concentration and DPPH free radical scavenging rate relation

From the Fig 6, the flavonoids scavenging rate of DPPH • increases first then decreases,reached minimum at 13μg/ml,the scavenging rate is 56.91%.The scavenging rate is better than V_C.

2.3.3 Determination of reducing power

The results were showed in Fig 7.From the Fig 7: Flavonoids and V_C reducing force decreased with the increasing of concentration then increased. Concentration at 7μg/ml flavonoids and V_C clearance gap was minimum, at 13μg/ml was maxmum. Flavonoids reducing force stronger than Vc.

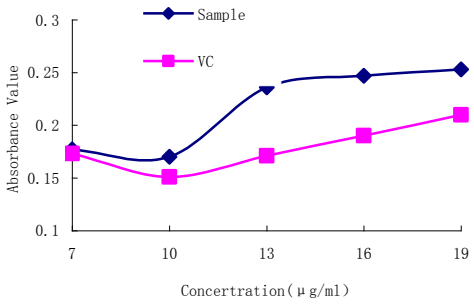


Fig.7 lavonoid concentration and reducing power

2.4 Experimental validation

Through the experiment,the optimum extraction conditions: liquid-solid ratio of 29.79:1, extraction temperature of 60.33℃, volume fraction of 69.60% ethanol solution, the yield of flavonoids is 41.9887mg /100 g. Considering the actual convenience of operation, the optimal process parameters for liquid ratio of 30:1, 70%

ethanol concentration, extraction temperature of 60℃. According to determine the optimum process conditions of extract ,repeated 3 times, the yield of flavonoids average is 41.0223 mg/100g, attain97.698% can better reflect the actual situation.

3.Conclusion

This test finally determine the best extraction process parameters : liquid-solid ratio of 30:1, extraction temperature of 60 ℃, 70% ethanol solution, the yield of flavonoid is 41.0223 mg/100 g; Antioxidant activity showed the results: black beans flavonoids had good antioxidant activity in vitro.

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